

Model Selection and Improvement Summary: Transfer Learning using ResNet

1. Overview

This report provides a summary of the model selection strategy and potential improvements performed to address overfitting for an image classification task (binary classification: cat or dog). The project employs transfer learning with a ResNet-based model and regularization techniques and fine-tuning strategies.

2. Model Selection Justification

The chosen model is a **pre-trained ResNet (Residual Network)** from the Keras applications library, integrated using **Transfer Learning**. The decision was made based on the following factors:

- **ResNet** has demonstrated high accuracy and robustness on large-scale image datasets such as ImageNet.
- It enables faster training by leveraging pre-trained convolutional filters that already capture general visual features.
- The **residual connections** in ResNet mitigate the vanishing gradient problem, allowing deeper networks to converge effectively.

Why Not Other Models

- **CNN from scratch:** Requires significantly more data and time to achieve comparable accuracy.
- **RNN:** Designed for sequential data (e.g., text, time series) and not suitable for image data.
- **Logistic Regression:** Too simplistic and incapable of handling complex image features.

3. Techniques to Prevent Overfitting

To improve generalization and address overfitting, the following regularization techniques were applied:

- **Dropout layers** (rate = 0.5) before and after the dense layer to randomly deactivate neurons during training.
- **L2 regularization** on the dense layer to penalize large weights.
- **Global Average Pooling** to reduce parameters and focus on spatially averaged features.
- **Fine-tuning:** Unfreezing the ResNet base and retraining with a low learning rate (0.0001) to adapt it to the new dataset.

4. Performance Summary

- **Validation Accuracy:** Achieved up to **97.9%**.
- **Classification Report:** High precision, recall, and F1-score with macro and weighted F1-scores of 0.97.
- **Overfitting Controlled:** Training and validation metrics aligned well, indicating stable generalization.

5. Suggested Improvement

With more time, the following improvements could be implemented:

- **Advanced Data Augmentation:** Techniques such as MixUp, CutMix, or random erasing.
- **Alternative Architectures:** Exploring EfficientNet or DenseNet for better efficiency and accuracy.
- **Hyperparameter Tuning:** Using tools like Keras Tuner or Optuna to optimize dropout rate, learning rate, etc.
- **Model Ensembling:** Combining predictions from multiple models for improved robustness.

6. Conclusion

The use of Transfer Learning with ResNet, supported by dropout, L2 regularization, and fine-tuning, resulted in a high-accuracy model with strong generalization. With more time, further enhancements such as advanced augmentation and hyperparameter tuning could provide additional performance improvements.

References

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